CCNA 200-301

Drag & Drop
Questions et réponses

32 exercices • 3 modules

Énoncés, éléments disponibles, associations correctes, schémas et explications. Les textes techniques restent en anglais pour conserver le contenu original des fichiers fournis.

Les réponses et les explications sont reprises des sources, y compris les questions qui présentent des réserves documentaires. Toutes les questions des trois modules sont incluses.

Module	Questions
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Sources : page HTML CCNA 200-301 Questions ; PDF Drag Drop Quiz 2 et Drag Drop Quiz 3. Le doublon Drag Drop Quiz 2 (1) ne répète pas les exercices.

Drag Drop Questions | Question 01

Refer to the exhibit.

```
[root#HostTime=]#ip route
default via 192.168.1.193 dev eth1 proto static
192.168.1.0/26 dev sth1 proto kernel scope link src 192.168.1.200 metric 1

[root#HostTime=]#ip addr show eth1
eth1:mtu 1500 qdisc pfifo_fast qlan 1000
 link/ether 00:0C:22:83:79:A3 brd ff:ff:ff:ff:ff:ff
 inet 192.168.1.200/26 hrd 192.168.1.255 scope global eth1
 inet6 fe80::20c::29ff:fe89:79b3/64 scope link
 valid_lft forever preferred_lft forever
```

Drag and drop the networking parameters from the left onto the correct values on the right.

Éléments de réponse disponibles

1. NIC vendor OUI
2. NIC MAC address
3. default gateway
4. host IP address
5. subnet mask

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
00:0C:22	NIC vendor OUI
00:0C:22:83:79:A3	NIC MAC address
192.168.1.193	default gateway
192.168.1.200	host IP address
255.255.255.192	subnet mask

Explication originale

+ 00:0C:22: NIC vendor OUI
+ 00:0C:22:83:79:A3: NIC MAC address
+ 192.168.1.193: default gateway
+ 192.168.1.200: host IP address
+ 255.255.255.192: subnet mask

The “ip route” and “ip addr show eth1” are Linux commands.
+ “ip route”: display the routing table
+ “ip addr show eth1”: get depth information (only on eth1 interface) about your network interfaces like IP Address, MAC Address information

Source : CCNA 200-301 Questions.html | question 1

Drag Drop Questions | Question 02

Drag and drop the functions from the left onto the correct network components on the right.

Éléments de réponse disponibles

1. DHCP Server

2. DNS Server

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
1. holds the TCP/IP settings to be distributed to the clients	DHCP Server
2. resolves web URLs to IP addresses	DNS Server
3. stores a list of IP addresses mapped to names	DNS Server
4. assigns a default gateway to a client	DHCP Server
5. assigns IP addresses to enabled clients	DHCP Server

Source : CCNA 200-301 Questions.html | question 2

Drag Drop Questions | Question 03

Drag the OSPF parameters to the correct places on the right. Note: There is one unused answer in this question.

Éléments de réponse disponibles

1. must be unique
2. must match
3. unused answer

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
1. netmask	must match
2. OSPF process ID	unused answer
3. router ID	must be unique
4. IP address	must be unique
5. area ID	must match
6. timers	must match

Source : CCNA 200-301 Questions.html | question 3

Drag Drop Questions | Question 04

Drag and drop the IPv4 network subnets from the left onto the correct usable host ranges on the right.

Éléments de réponse disponibles

1. 172.28.192.1 - 172.28.255.254
2. 172.28.224.1 - 172.28.231.254
3. 172.28.228.1 - 172.28.229.254
4. 172.28.228.129 - 172.28.228.254
5. 172.28.228.145 - 172.28.228.150

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
172.28.228.144/18	172.28.192.1 - 172.28.255.254
172.28.228.144/21	172.28.224.1 - 172.28.231.254
172.28.228.144/23	172.28.228.1 - 172.28.229.254
172.28.228.144/25	172.28.228.129 - 172.28.228.254
172.28.228.144/29	172.28.228.145 - 172.28.228.150

Explication originale

+ 172.28.228.144/18: 172.28.192.1 - 172.28.255.254
+ 172.28.228.144/21: 172.28.224.1 - 172.28.231.254
+ 172.28.228.144/23: 172.28.228.1 - 172.28.229.254
+ 172.28.228.144/25: 172.28.228.129 - 172.28.228.254
+ 172.28.228.144/29: 172.28.228.145 - 172.28.228.150

This subnet question requires us to grasp how to subnet very well. To quickly find out the subnet range, we have to find out the increment and the network address of each subnet. Let’s take an example with the subnet 172.28.228.144/18:

From the /18 (= 1100 0000 in the 3rd octet), we find out the increment is 64. Therefore the network address of this subnet must be the greatest multiple of the increment but not greater than the value in the 3rd octet (228). We can find out the 3rd octet of the network address is 192 (because 192 = 64 * 3 and 192 < 228) -> The network address is 172.28.192.0. So the first usable host should be 172.28.192.1 and it matches with the 5th answer on the right. In this case we don’t need to calculate the broadcast address because we found the correct answer.

Let’s take another example with subnet 172.28.228.144/23 -> The increment is 2 (as /23 = 1111 1110 in 3rd octet). The 3rd octet of the network address is 228 (because 228 is the multiply of 2 and equal to the 3rd octet) -> The network address is 172.28.228.0 -> The first usable host is 172.28.228.1. It is not necessary but if we want to find out the broadcast address of this subnet, we can find out the next network address, which is 172.28.(228 + the increment number).0 or 172.28.230.0 then reduce 1 bit -> 172.28.229.255 is the broadcast address of our subnet. Therefore the last usable host is 172.28.229.254.

If you are still unclear about how to do subnetting quickly, please read part 3 of our “Subnetting Tutorial - Subnetting Made Easy” at https://www.9tut.com/subnetting-tutorial/3

Source : CCNA 200-301 Questions.html | question 4

Drag Drop Questions | Question 05

Drag and drop the threat-mitigation techniques from the left onto the types of threat or attack they mitigate on the right.

Éléments de réponse disponibles

1. Configure a VLAN access control list
2. Configure the dynamic ARP inspection feature
3. Configure the root guard feature
4. Configure the BPDU guard feature

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
802.1q double tagging	Configure a VLAN access control list
ARP spoofing	Configure the dynamic ARP inspection feature
unwanted superior BPDUs	Configure the root guard feature
unwanted BPDUs on PortFast-enabled interfaces	Configure the BPDU guard feature

Explication originale

+ 802.1q double tagging: Configure VACL
+ ARP spoofing: Configure dynamic ARP inspection
+ unwanted superior BPDUs: Configure root guard
+ unwanted BPDUs on PortFast-enabled interfaces: Configure BPDU guard

Double-Tagging attack:

Attacker --VLAN 10--> Company Switch A --Native VLAN 10--> Company Switch B --VLAN 20--> Victim [Payload|VLAN 20|VLAN 10] [Payload|VLAN 20] [Payload]

In this attack, the attacking computer generates frames with two 802.1Q tags. The first tag matches the native VLAN of the trunk port (VLAN 10 in this case), and the second matches the VLAN of a host it wants to attack (VLAN 20).

When the packet from the attacker reaches Switch A, Switch A only sees the first VLAN 10 and it matches with its native VLAN 10 so this VLAN tag is removed. Switch A forwards the frame out all links with the same native VLAN 10. Switch B receives the frame with an tag of VLAN 20 so it removes this tag and forwards out to the Victim computer.

Note: This attack only works if the trunk (between two switches) has the same native VLAN as the attacker.

To mitigate this type of attack, you can use VLAN access control lists (VACLs, which applies to all traffic within a VLAN. We can use VACL to drop attacker traffic to specific victims/servers) or implement Private VLANs.

ARP attack (like ARP poisoning/spoofing) is a type of attack in which a malicious actor sends falsified ARP messages over a local area network as ARP allows a gratuitous reply from a host even if an ARP request was not received. This results in the linking of an attacker’s MAC address with the IP address of a legitimate computer or server on the network. This is an attack based on ARP which is at Layer 2. Dynamic ARP inspection (DAI) is a security feature that validates ARP packets in a network which can be used to mitigate this type of attack.

Source : CCNA 200-301 Questions.html | question 5

Drag Drop Questions | Question 06

Drag and drop the description of file-transfer protocols from the left onto the correct protocols on the right.

Éléments de réponse disponibles

1. FTP

2. TFTP

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
1. does not require user authentication	TFTP
2. uses UDP	TFTP
3. uses TCP	FTP
4. uses ports 20 and 21	FTP
5. provides reliability when loading an IOS image upon boot up	FTP
6. uses port 69	TFTP

Source : CCNA 200-301 Questions.html | question 6

Drag Drop Questions | Question 07

Drag and drop the application protocols from the left onto the suitable transport protocols on the right.

Éléments de réponse disponibles

1. TCP

2. UDP

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
1. SMTP	TCP
2. SNMP	UDP
3. DHCP	UDP
4. TFTP	UDP
5. FTP	TCP
6. SSH	TCP

Source : CCNA 200-301 Questions.html | question 7

Drag Drop Questions | Question 08

Drag and drop the descriptions of IP protocol transmissions from the left onto the correct IP traffic types on the right.

Éléments de réponse disponibles

1. TCP

2. UDP

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
1. Sends Transmission in Sequence	TCP
2. Transmits packets individually	UDP
3. Uses higher transmission rate to support latency-sensitive apps	UDP
4. Uses a lower transmission rate to ensure reliability	TCP
5. Transmissions include an 8-byte header	UDP
6. Transmits packet as stream	TCP

Explication originale

UDP is a simpler message-based connectionless protocol. In connectionless protocols, there is no effort made to setup a dedicated end-to-end connection.

...

Datagrams - Packets are sent individually and are guaranteed to be whole if they arrive. Packets have definite bounds and no split or merge into data streams may exist.

According to https://community.cisco.com/t5/networking-documents/udp/ta-p/3114870

TCP is connection-oriented and UDP is connectionless. This means that before sending TCP packets, a connection is established between the server and the client. This process of setting up a connection is called TCP handshaking. The stream of packets is then sent over this connection -> TCP transmits packet as stream.

Reference: https://www.vpnmentor.com/blog/tcp-vs-udp/

Source : CCNA 200-301 Questions.html | question 8

Drag Drop Questions | Question 09

Drag and drop the WLAN components from the left onto the correct descriptions on the right.

Éléments de réponse disponibles

1. wireless LAN controller
2. access point
3. service port
4. virtual interface
5. dynamic interface

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
device that manages access points	wireless LAN controller
device that provides Wi-Fi devices with a connection to a wired network	access point
used for out of band management of a WLC	service port
used to support mobility management of the WLC	virtual interface
applied to the WLAN for wireless client communication	dynamic interface

Explication originale

+ device that manages access points: wireless LAN controller
+ device that provides Wi-Fi devices with a connection to a wired network: access point
+ used for out of band management of a WLC: service port
+ used to support mobility management of the WLC: virtual interface
+ applied to the WLAN for wireless client communication: dynamic interface

The service port can be used management purposes, primarily for out-of-band management. However, AP management traffic is not possible across the service port. In most cases, the service port is used as a “last resort” means of accessing the controller GUI for management purposes. For example, in the case where the system distribution ports on the controller are down or their communication to the wired network is otherwise degraded.

A dynamic interface with the Dynamic AP Management option enabled is used as the tunnel source for packets from the controller to the access point and as the destination for CAPWAP packets from the access point to the controller.

The virtual interface is used to support mobility management, Dynamic Host Configuration Protocol (DHCP) relay, and embedded Layer 3 security such as guest web authentication. It also maintains the DNS gateway host name used by Layer 3 security and mobility managers to verify the source of certificates when Layer 3 web authorization is enabled.

Reference:
https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-5/config-guide/b_cg85/ports_and_interfaces.html

Source : CCNA 200-301 Questions.html | question 9

Drag Drop Questions | Question 10

Drag and drop the networking parameters from the left on to the correct values on the right.

Éléments de réponse disponibles

1. Connection Oriented

2. Connectionless

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
1. SMTP	Connection Oriented
2. SNMP	Connectionless
3. TFTP	Connectionless
4. VoIP	Connectionless
5. SSH	Connection Oriented
6. FTP	Connection Oriented

Explication originale

SSH uses TCP port 22 while SNMP uses UDP port 161 and 162.

Source : CCNA 200-301 Questions.html | question 10

Drag Drop Questions | Question 11

Drag drop the descriptions from the left on to the correct configuration-management technologies on the right.

Éléments de réponse disponibles

1. Ansible
2. Chef
3. Puppet

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
1. uses Ruby for fundamental configuration elements	Chef
2. uses TCP port 10002 for configuration push jobs	Chef
3. uses SSH for remote device communication	Ansible
4. fundamental configuration elements are stored in a manifest	Puppet
5. uses TCP 8140 for communication	Puppet
6. uses YAML for fundamental configuration elements	Ansible

Explication originale

The focus of Ansible is to be streamlined and fast, and to require no node agent installation. Thus, Ansible performs all functions over SSH. Ansible is built on Python, in contrast to the Ruby foundation of Puppet and Chef.

TCP port 10002 is the command port. It may be configured in the Chef Push Jobs configuration file . This port allows Chef Push Jobs clients to communicate with the Chef Push Jobs server.

Puppet is an open-source configuration management solution, which is built with Ruby and offers custom Domain Specific Language (DSL) and Embedded Ruby (ERB) templates to create custom Puppet language files, offering a declarative-paradigm programming approach.

A Puppet piece of code is called a manifest, and is a file with .pp extension.

Source : CCNA 200-301 Questions.html | question 11

Drag Drop Questions | Question 12

Drag and drop the AAA functions from the left onto the correct AAA services on the right.

Éléments de réponse disponibles

1. Authentication
2. Authorization
3. Accounting

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
1. records user activities	Accounting
2. restricts the services that are available to a user	Authorization
3. identifies the user	Authentication
4. controls the actions that a user can perform	Authorization
5. provides analytical information for the network administrator	Accounting
6. verifies the password associated with a user	Authentication

Source : CCNA 200-301 Questions.html | question 12

Drag Drop Questions | Question 13

Drag and Drop the benefits of a Cisco Wireless LAN Controller from the left onto the correct examples on the right.

Éléments de réponse disponibles

1. Access points auto adjust signal strength
2. Controller provides centralized management of users and VLANs
3. Controller uses loadbalancing to maximize throughput
4. Controller image auto deployed to access Points

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
Dynamic RF Feature	Access points auto adjust signal strength
Easy Deployment Process	Controller provides centralized management of users and VLANs
Optimized user performance	Controller uses loadbalancing to maximize throughput
Easy upgrade process	Controller image auto deployed to access Points

Explication originale

+ Dynamic RF Feature: Access points auto adjust signal strength
+ Easy Deployment Process: Controller provides centralized management of users and VLANs
+ Optimized user performance: Controller uses loadbalancing to maximize throughput
+ Easy upgrade process: Controller image auto deployed to access Points

Source : CCNA 200-301 Questions.html | question 13

Drag Drop Questions | Question 14

Drag and drop the Cisco Wireless LAN Controller security settings from the left onto the correct security mechanism categories on the right.

Éléments de réponse disponibles

1. Layer 2 Security Mechanism

2. Layer 3 Security Mechanisms (for WLAN)

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
1. web policy	Layer 3 Security Mechanisms (for WLAN)
2. Passthrough	Layer 3 Security Mechanisms (for WLAN)
3. WPA+WPA2	Layer 2 Security Mechanism
4. 802.1X	Layer 2 Security Mechanism

Explication originale

Layer 2 Security Mechanism includes WPA+WPA2, 802.1X, Static WEP, CKIP while Layer 3 Security Mechanisms (for WLAN) includes IPSec, VPN Pass-Through, Web Passthrough ...

Reference: https://www.cisco.com/c/en/us/support/docs/wireless/4400-series-wireless-lan-controllers/106082-wlc-compatibility-matrix.html

Source : CCNA 200-301 Questions.html | question 14

Drag Drop Questions 2 | Question 01

Drag each route source from the left to the numbers on the right. Beginning with the lowest and ending with the highest administrative distance.

Éléments de réponse disponibles

1. connected
2. static
3. EBGP
4. EIGRP
5. OSPF
6. RIP

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
1	connected
2	static
3	EBGP
4	EIGRP
5	OSPF
6	RIP

Explication originale

1: connected (AD = 0)
2: static (AD = 1)
3: EBGP (AD = 20)
4: EIGRP (AD = 90)
5: OSPF (AD = 110)
6: RIP (AD = 120)

Source : CCNA Training » Drag Drop Quiz 2.pdf | question 1 | page 1

Drag Drop Questions 2 | Question 02

Order the DHCP message types as they would occur between a DHCP client and a DHCP server.

Éléments de réponse disponibles

1. DHCPDISCOVER
2. DHCPOFFER
3. DHCPREQUEST
4. DHCPACK

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
First	DHCPDISCOVER
Second	DHCPOFFER
Third	DHCPREQUEST
Fourth	DHCPACK

Explication originale

There are four messages sent between the DHCP Client and DHCP Server: DHCPDISCOVER, DHCPOFFER, DHCPREQUEST and DHCPACKNOWLEDGEMENT. This process is often abbreviated as DORA (for Discover, Offer, Request, Acknowledgement).
Answer:
+ First: DHCPDISCOVER
+ Second: DHCPOFFER
+ Third: DHCPREQUEST
+ Fourth: DHCPACK

Source : CCNA Training » Drag Drop Quiz 2.pdf | question 2 | page 2

Drag Drop Questions 2 | Question 03

An interface has been configured with the access list that is shown below.

access-list 107 deny tcp 207.16.12.0 0.0.3.255 any eq http
access-list 107 permit ip any any

On the basis of that access list, drag each information packet on the left to the appropriate category on the right.

Éléments de réponse disponibles

1. Permitted

2. Denied

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
source IP: 207.16.32.14, destination application: http	Permitted
source IP: 207.16.15.9, destination port: 23	Permitted
source IP: 207.16.14.7, destination port: 80	Denied
source IP: 207.16.13.14, destination application: http	Denied
source IP: 207.16.16.14, destination port: 53	Permitted

Explication originale

From the ACL:
access-list 107 deny tcp 207.16.12.0 0.0.3.255 any eq http
access-list 107 permit ip any any
We can deduce it block all source hosts in 207.16.12.0/22 range (which ranges from 207.16.12.0 to 207.16.15.255) via HTTP (port 80) while permit any other hosts.

Source : CCNA Training » Drag Drop Quiz 2.pdf | question 3 | page 2

Drag Drop Questions 2 | Question 04

Match the functions to the corresponding layers. (Not all options are used)

Éléments de réponse disponibles

1. Access layer
2. Distribution layer
3. Core layer
4. Unused

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
provides high-speed backbone connectivity	Core layer
implements personal firewalls on the client computers	Unused
provides network access to the user	Access layer
implements network access policy	Distribution layer
represents the network edge	Access layer
establishes Layer 3 routing boundaries	Distribution layer
functions as an aggregator for all the campus blocks	Core layer

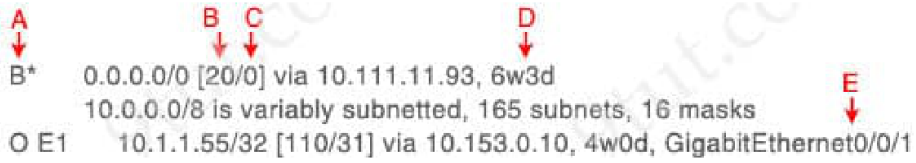
Explication originale

A typical enterprise hierarchical LAN campus network design includes the following three layers:
+ Access layer: Provides workgroup/user access to the network
+ Distribution layer: Provides policy-based connectivity and controls the boundary between the access and core layers
+ Core layer: Provides fast transport between distribution switches within the enterprise campus
Reference: https://www.ciscopress.com/articles/article.asp?p=2202410&seqNum=4

Source : CCNA Training » Drag Drop Quiz 2.pdf | question 4 | page 3

Drag Drop Questions 2 | Question 05

Refer to the exhibit. Drag and drop the routing table components on the left onto the corresponding letter from the exhibit on the right. Not all options are used.



Éléments de réponse disponibles

1. administrative distance
2. metric
3. next-hop interface
4. outbound interface
5. route source
6. subnet mask
7. timestamp

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
A	route source
B	administrative distance
C	metric
D	timestamp
E	outbound interface

Éléments non utilisés : next-hop interface; subnet mask

Source : CCNA Training » Drag Drop Quiz 2.pdf | question 5 | page 4

Drag Drop Questions 2 | Question 06

Drag the descriptions of device management from the left onto the types of device management on the right.

Éléments de réponse disponibles

1. Cisco DNA Center Device Management
2. Traditional Device Management

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
monitors the cloud for software updates	Cisco DNA Center Device Management
manages device configurations on a per-device basis	Traditional Device Management
implements changes via an SSH terminal	Traditional Device Management
uses CLI templates to apply a consistent configuration to multiple devices at an individual location	Cisco DNA Center Device Management
security is managed near the perimeter of the network with firewalls, VPNs and IPS	Traditional Device Management
uses NetFlow to analyze potential security threats throughout the network and take appropriate action on that traffic	Cisco DNA Center Device Management

Source : CCNA Training » Drag Drop Quiz 2.pdf | question 6 | page 4

Drag Drop Questions 2 | Question 07

Drag and drop each broadcast IP address on the left to the Broadcast Address column on the right. Not all options are used.

Éléments de réponse disponibles

1. Broadcast address
2. Unused

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
10.1.255.254/24	Unused
10.63.255.255/10	Broadcast address
172.16.255.39/29	Broadcast address
172.20.255.255/16	Broadcast address
192.168.1.10/24	Unused
192.168.255.127/25	Broadcast address

Source : CCNA Training » Drag Drop Quiz 2.pdf | question 7 | page 5

Drag Drop Questions 2 | Question 08

Drag and drop to the characteristics of networking from the left onto the correct networking types on the right.

Éléments de réponse disponibles

1. Controller-Based Networking
2. Traditional Networking

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
focused on network	Controller-Based Networking
focused on devices	Traditional Networking
user input is a configuration	Traditional Networking
user input is a policy	Controller-Based Networking
uses allow list security model	Controller-Based Networking
uses block list security model	Traditional Networking

Explication originale

Traditional Networking uses block list model because all traffic is allowed to pass through initially until we configure an ACL, prefix-list... This model is more lenient, because if no application knowledge exists, the default network configuration will let everything through. From a security perspective, it is very dangerous though - every time you allow a protocol that the application does not need, you are unnecessarily increasing your attack surface, and therefore compromising the overall application security.
In Controller-Based Networking, everything is denied by default until we explicitly allow them by a white (allow) list.

Source : CCNA Training » Drag Drop Quiz 2.pdf | question 8 | page 6

Drag Drop Questions 2 | Question 09

Drag and drop the attack-mitigation techniques from the left onto the types of attack that they mitigate on the right.

Éléments de réponse disponibles

1. configure the native VLAN with a nondefault VLAN ID
2. configure 802.1x authenticate
3. configure DHCP snooping
4. disable DTP

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
802.1q double-tagging VLAN-hopping attack	configure the native VLAN with a nondefault VLAN ID
MAC flooding attack	configure 802.1x authenticate
man-in-the-middle spoofing attack	configure DHCP snooping
switch-spoofing VLAN-hopping attack	disable DTP

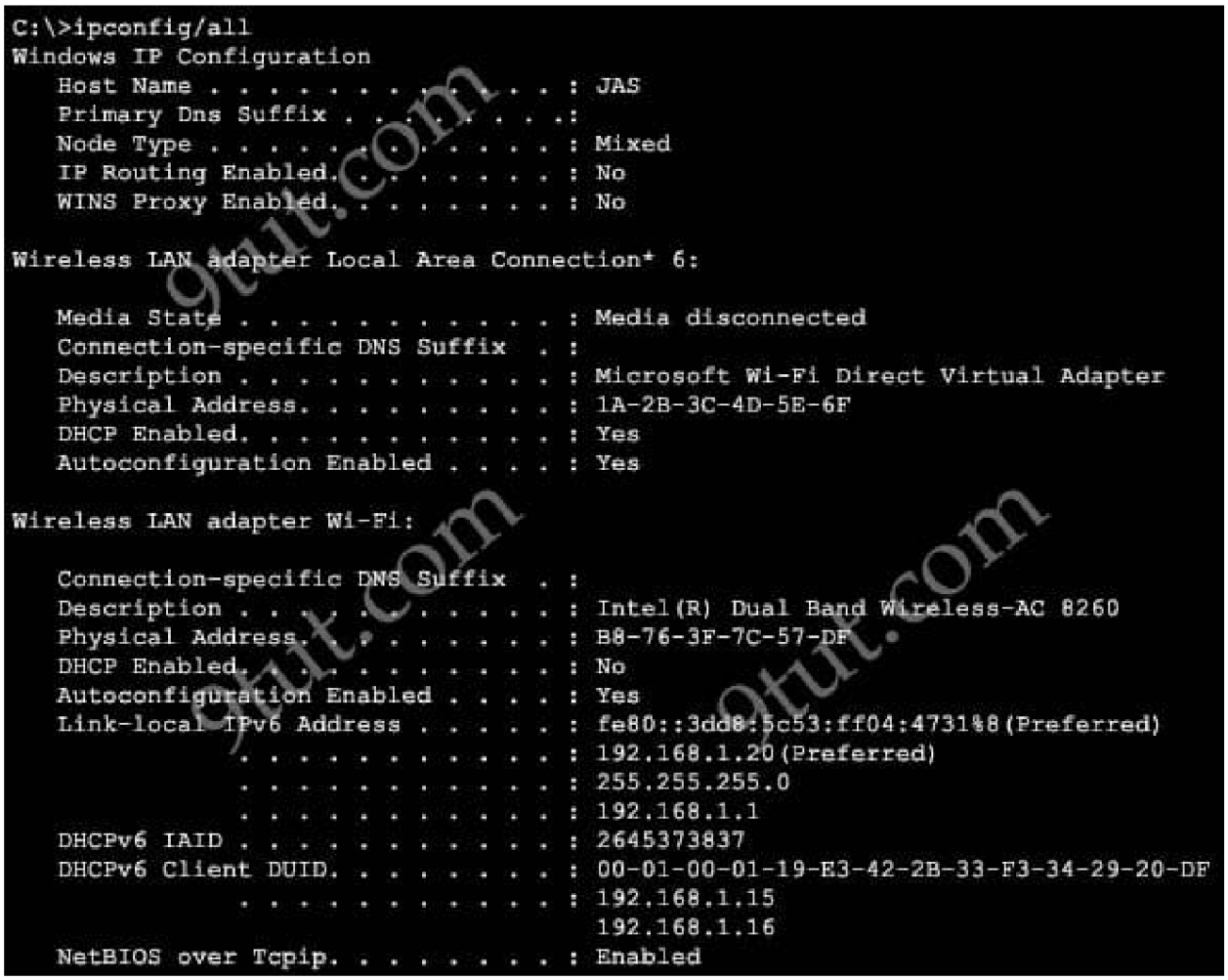
Explication originale

VLAN Hopping: By altering the VLAN ID on packets encapsulated for trunking, an attacking device can send or receive packets on various VLANs, bypassing Layer 3 security measures. VLAN hopping can be accomplished by switch spoofing or double tagging.
a. Switch spoofing:
The attacker can connect an unauthorized Cisco switch to a Company switch port. The unauthorized switch can send DTP frames and form a trunk with the Company Switch. If the attacker can establish a trunk link to the Company switch, it receives traffic to all VLANs through the trunk because all VLANs are allowed on a trunk by default.
(Instead of using a Cisco Switch, the attacker can use a software to create and send DTP frames).
To mitigate this type of attack, we can disable DTP.
b. Double-Tagging attack:
In this attack, the attacking computer generates frames with two 802.1Q tags. The first tag matches the native VLAN of the trunk port (VLAN 10 in this case), and the second matches the VLAN of a host it wants to attack (VLAN 20).
When the packet from the attacker reaches Switch A, Switch answer " only sees the first VLAN 10 and it matches with its native VLAN 10 so this VLAN tag is removed. Switch answer " forwards the frame out all links with the same native VLAN 10. Switch answer " receives the frame with an tag of VLAN 20 so it removes this tag and forwards out to the Victim computer.
Note: This attack only works if the trunk (between two switches) has the same native VLAN as the attacker.
To mitigate this type of attack, we can use VLAN access control lists (VACLs, which applies to all traffic within a VLAN. We can use VACL to drop attacker traffic to specific victims/servers) or implement Private VLANs.
Answer:
+ 802.1q double-tagging VLAN-hopping attack: configure the native VLAN with a nondefault VLAN ID
+ MAC flooding attack: configure 802.1x authenticate
+ man-in-the-middle spoofing attack: configure DHCP snooping
+ switch-spoofing VLAN-hopping attack: disable DTP

Source : CCNA Training » Drag Drop Quiz 2.pdf | question 9 | page 6

Drag Drop Questions 2 | Question 10

An engineer is tasked with verifying network configuration parameters on a client workstation to report back to the team lead. Drag and drop the node identifiers from the left onto the network parameters on the right.



Éléments de réponse disponibles

1. 192.168.1.255
2. 192.168.1.1
3. 192.168.1.20
4. B8-76-3F-7C-57-DF
5. 192.168.1.254

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
broadcast address	192.168.1.255
default gateway	192.168.1.1
host IP address	192.168.1.20
MAC address	B8-76-3F-7C-57-DF
last assignable IP address in the subnet	192.168.1.254

Explication originale

In the output above, three lines under "Link-local IPv6 Address" line have been blacked out so we have to figure out what they are. They are the IP address of the Wireless card and the Default gateway of this computer.
Answer:
+ broadcast address: 192.168.1.255
+ default gateway: 192.168.1.1
+ host IP address: 192.168.1.20
+ MAC address: B8-76-3F-7C-57-DF
+ last assignable IP address in the subnet: 192.168.1.254

Source : CCNA Training » Drag Drop Quiz 2.pdf | question 10 | page 8

Drag Drop Questions 3 | Question 01

Refer to the exhibit. An engineer is required to verify that the network parameters are valid for the users wireless LAN connectivity on a /24 subnet. Drag and drop the values from the left onto the network parameters on the right. Not all values are used.



Éléments de réponse disponibles

1. 192.168.1.20
2. 192.168.1.255
3. 192.168.1.0
4. 192.168.1.1
5. B8-76-3F-7C-57-DF
6. 192.168.1.254
7. 1A-76-3F-7C-57-DF

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
host IP address	192.168.1.20
broadcast address	192.168.1.255
network address	192.168.1.0
default gateway	192.168.1.1
MAC address	B8-76-3F-7C-57-DF
last assignable IP address in the subnet	192.168.1.254
Unused	1A-76-3F-7C-57-DF

Explication originale

+ host IP address: 192.168.1.20
+ broadcast address: 192.168.1.255
+ network address: 192.168.1.0
+ default gateway: 192.168.1.1
+ MAC address: B8-76-3F-7C-57-DF
+ last assignable IP address in the subnet: 192.168.1.254

Source : CCNA Training » Drag Drop Quiz 3.pdf | question 1 | page 1

Drag Drop Questions 3 | Question 02

Drag and drop the SNMP components from the left onto the descriptions on the right.

Éléments de réponse disponibles

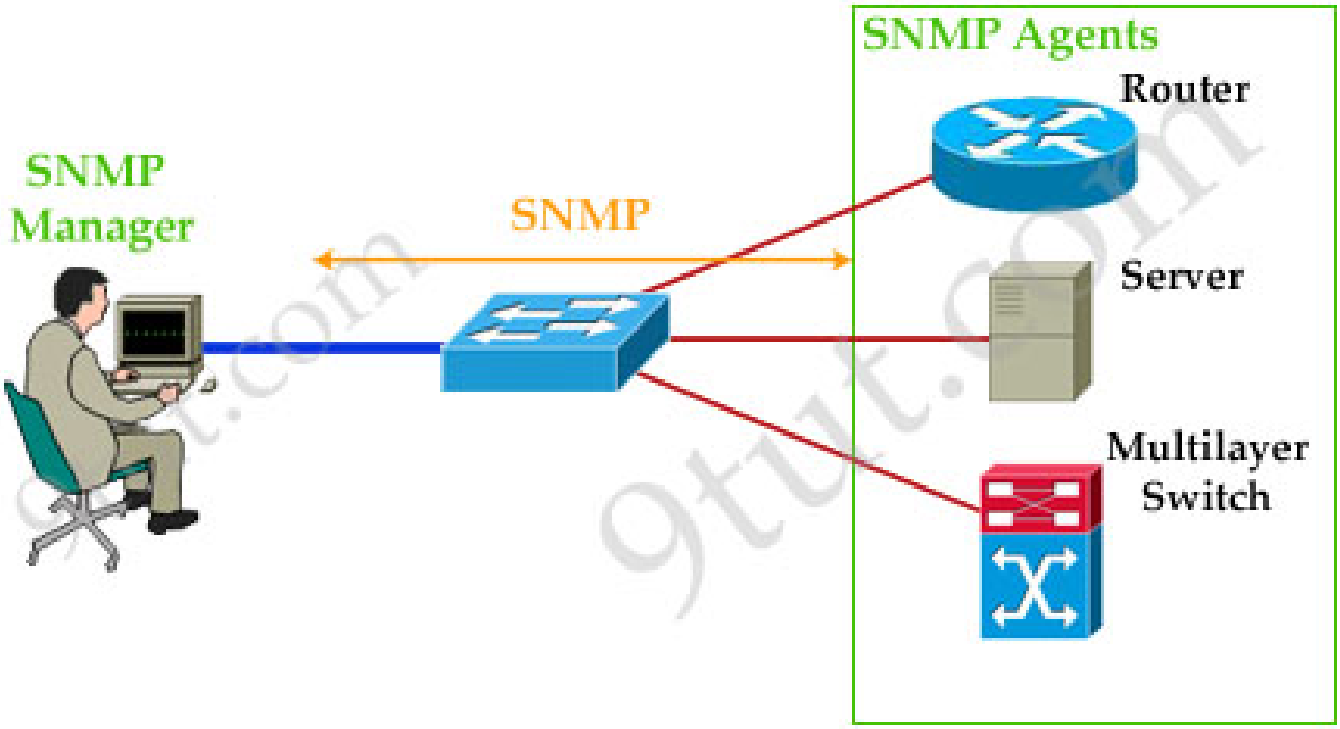
1. SNMP trap
2. SNMP agent
3. MIB
4. SNMP manager

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
unsolicited message	SNMP trap
responds to status requests and requests for information about a device	SNMP agent
collection of variables that can be monitored	MIB
resides on an NMS	SNMP manager

Explication originale

+ unsolicited message: SNMP trap
+ responds to status requests and requests for information about a device: SNMP agent
+ collection of variables that can be monitored: MIB
+ resides on an NMS: SNMP manager
SNMP consists of 3 items:
+ SNMP Manager (sometimes called Network Management System - NMS): a software runs on the device of the network administrator (in most case, a computer) to monitor the network.
+ SNMP Agent: a software runs on network devices that we want to monitor (router, switch, server…)
+ Management Information Base (MIB): is the collection of managed objects. This components makes sure that the data exchange between the manager and the agent remains structured. In other words, MIB contains a
set of questions that the SNMP Manager can ask the Agent (and the Agent can understand them). MIB is commonly shared between the Agent and Manager.



Source : CCNA Training » Drag Drop Quiz 3.pdf | question 2 | page 3

Drag Drop Questions 3 | Question 03

Drag and drop the QoS congestion management terms from the left onto the description on the right.

Éléments de réponse disponibles

1. CQ
2. CBWFQ
3. PQ
4. WFQ
5. FIFO

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
services a specified number of bytes in one queue before continuing to the next queue	CQ
provides guaranteed bandwidth to a specified class of traffic	CBWFQ
places packets into one of four priority-based queues	PQ
provides minimum guaranteed bandwidth to one or more flows	WFQ
uses store-and-forward queuing	FIFO

Explication originale

+ services a specified number of bytes in one queue before continuing to the next queue: CQ
+ provides guaranteed bandwidth to a specified class of traffic: CBWFQ
+ places packets into one of four priority-based queues: PQ
+ provides minimum guaranteed bandwidth to one or more flows: WFQ
+ uses store-and-forward queuing: FIFO

First-in, first-out (FIFO): FIFO entails no concept of priority or classes of traffic. With FIFO, transmission of packets out the interface occurs in the order the packets arrive, which means no QoS
Priority Queuing (PQ): schedules traffic such that the higher-priority queues “always” get serviced first
Custom Queuing (CQ): provide specific traffic guaranteed bandwidth at a potential congestion point, assuring the traffic a fixed portion of available bandwidth and leaving the remaining bandwidth to other traffic.
Weighted fair queueing (WFQ): offers dynamic, fair queuing that divides bandwidth across queues of traffic based on weights. In standard WFQ, packets are classified into flows according to one of four criteria: the source Internet Protocol address (IP address), the destination IP address, the source Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) port, or the destination TCP or UDP port.
Class-based weighted fair queueing (CBWFQ) extends the standard WFQ functionality to provide support for user-defined traffic classes. For CBWFQ, you define traffic classes based on match criteria including protocols, access control lists (ACLs), and input interfaces. Packets satisfying the match criteria for a class constitute the traffic for that class. A queue is reserved for each class, and traffic belonging to a class is directed to the queue
for that class.
Once a class has been defined according to its match criteria, you can assign it characteristics. To characterize a
class, you assign it bandwidth, weight, and maximum packet limit. The bandwidth assigned to a class is the guaranteed bandwidth delivered to the class during congestion.

Source : CCNA Training » Drag Drop Quiz 3.pdf | question 3 | page 4

Drag Drop Questions 3 | Question 04

An engineer is tasked to configure a switch with port security to ensure devices that forward unicasts, multicasts, and broadcasts are unable to flood the port. The port must be configured to permit only two random MAC addresses at a time. Drag and drop the required configuration commands from the left onto the sequence on the right. (Not all commands are used)

Éléments de réponse disponibles

1. `switchport mode access`
2. `switchport port-security`
3. `switchport port-security mac-address 0060.3EDD.77ab`
4. `switchport port-security mac-address 00D0.D443.4564`
5. `switchport port-security mac-address sticky`
6. `switchport port-security violation shutdown`
7. `switchport port-security maximum 2`

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
1	switchport mode access
2	switchport port-security
3	switchport port-security maximum 2
4	switchport port-security violation shutdown

Éléments non utilisés : switchport port-security mac-address 0060.3EDD.77ab; switchport port-security mac-address 00D0.D443.4564; switchport port-security mac-address sticky

Explication originale

This question says "permit only two random MAC addresses at a time" so we cannot use the "switchport port-security mac-address sticky" command as the switch will only learn and save the first two MAC addresses forever. Although the command "switchport port-security violation shutdown" is unnecessary because it is the default state of port-security but it is not wrong to reuse it.
Note:
The full syntax of the second command is:
switchport port-security mac-address sticky [MAC]
If we don’t specify the MAC address (like in this question) then the switch will dynamically learn the attached MAC Address and place it into your running-configuration.

Source : CCNA Training » Drag Drop Quiz 3.pdf | question 4 | page 5

Drag Drop Questions 3 | Question 05

Drag the characteristics of network architectures from the left onto the type of architecture on the right.

Éléments de réponse disponibles

1. Collapsed Core

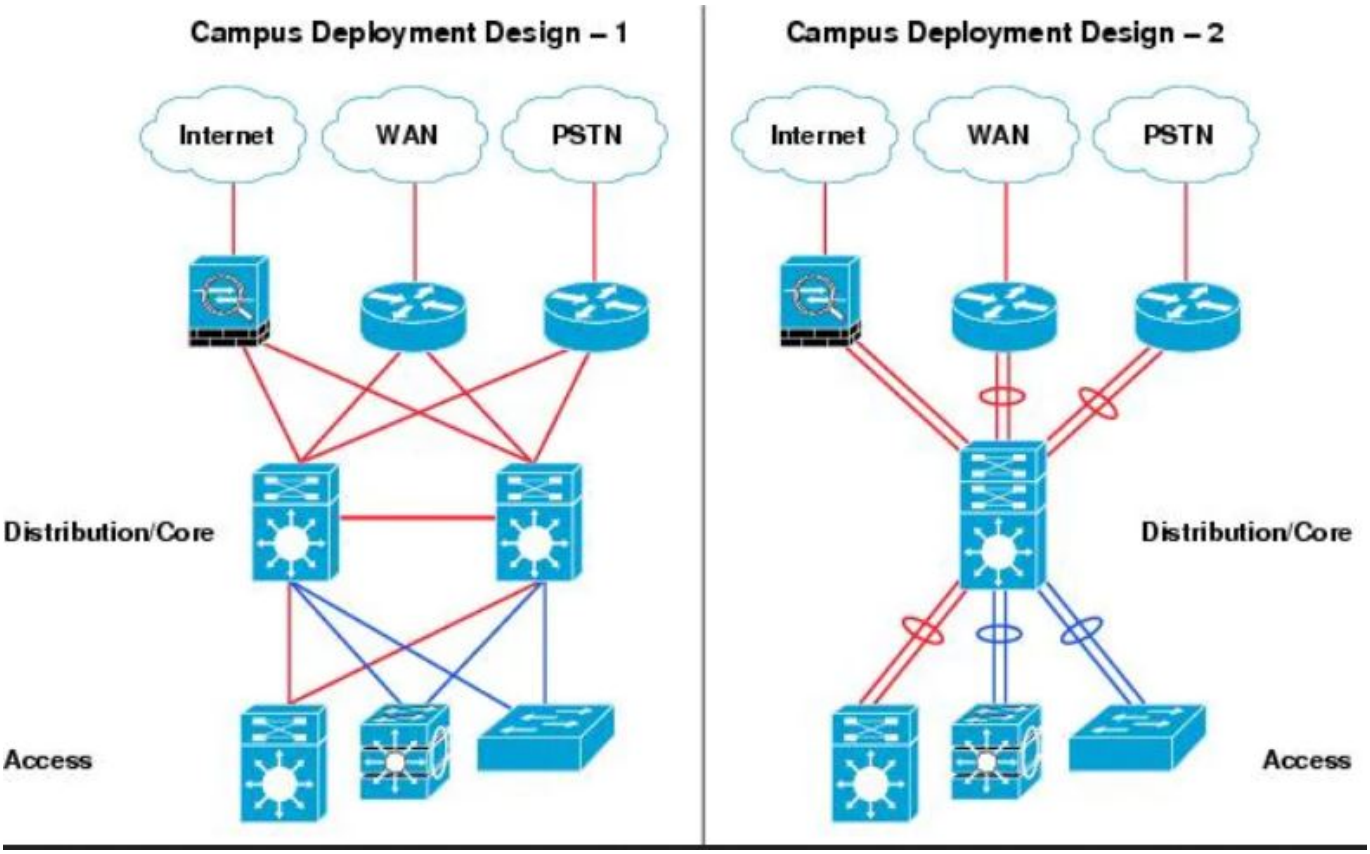
2. Three-Tier

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
most appropriate for small network designs	Collapsed Core
single device handles the core and the distribution layer	Collapsed Core
separate devices handle the core and the distribution layer	Three-Tier
more cost-effective than other options	Collapsed Core
enhances network availability	Three-Tier

Explication originale

The three-tier hierarchical design maximizes performance, network availability, and the ability to scale the network design.
However, many small enterprise networks do not grow significantly larger over time. Therefore, a two-tier hierarchical design where the core and distribution layers are collapsed into one layer is often more practical. A
“collapsed core” is when the distribution layer and core layer functions are implemented by a single device. The primary motivation for the collapsed core design is reducing network cost, while maintaining most of the benefits of the three-tier hierarchical model.
Reference: https://www.ciscopress.com/articles/article.asp?p=2202410&seqNum=4
A collapsed core network is shown below. The collapsed core network may be deployed with redundant core/distribution router, or consolidated core/distribution router.



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Drag Drop Questions 3 | Question 06

Refer to the exhibit. An engineer is configuring the router to provide static NAT for the webserver. Drag and drop the configuration commands from the left onto the letters that correspond to its position in the configuration on the right.



Webserver

172.16.1.2

interface Ethernet0

A

B

interface Serial0

C

D

E

F

access-list 1 permit 172.16.1.0 0.0.0.255

Éléments de réponse disponibles

1. ip address 172.16.1.1 255.255.255.0
2. ip nat inside
3. ip address 45.83.2.214 255.255.255.240
4. ip nat outside
5. ip nat inside source static tcp 172.16.1.2 80 45.83.2.214 80 extendable
6. ip nat inside source list 1 interface s0 overload

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
position A	ip address 172.16.1.1 255.255.255.0
position B	ip nat inside
position C	ip address 45.83.2.214 255.255.255.240
position D	ip nat outside
position E	ip nat inside source static tcp 172.16.1.2 80 45.83.2.214 80 extendable
position F	ip nat inside source list 1 interface s0 overload

Explication originale

+ position A: ip address 172.16.1.1 255.255.255.0
+ position B: ip nat inside
+ position C: ip address 45.83.2.214 255.255.255.240
+ position D: ip nat outside
+ position E: ip nat inside source static tcp 172.16.1.2 80 45.83.2.214 80 extendable
+ position F: ip nat inside source list 1 interface s0 overload

Source : CCNA Training » Drag Drop Quiz 3.pdf | question 6 | page 7

Drag Drop Questions 3 | Question 07

Drag the IPv6 DNS record types from the left onto the description on the right.

Éléments de réponse disponibles

1. NS
2. SOA
3. CNAME
4. PTR
5. AAAA

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
correlates a domain with its authoritative name servers	NS
associates the domain serial number with its owner	SOA
aliases one name to another	CNAME
supports reverse name lookups	PTR
correlates a host name with an IP address	AAAA

Explication originale

+ correlates a domain with its authoritative name servers: NS
+ associates the domain serial number with its owner: SOA
+ aliases one name to another: CNAME
+ supports reverse name lookups: PTR
+ correlates a host name with an IP address: AAAA
An AAAA record maps a domain name to the IP address (Version 6) of the computer hosting the domain. An AAAA record is used to find the IP address of a computer connected to the internet from a name. The AAAA record is conceptually similar to the A record, but it allows you to specify the IPv6 address of the server, rather than the IPv4. An example of AAAA is the ‘www’ of 9tut.com is pointed to
2001:0db8:aaaa:bbbb:cccc:dddd:eeee:ffff.
NS (name server) looks like a website URL, but instead of linking to a website, they link to the domain names authoritative nameservers. An example of a NS is ‘ns1.9tut.com’.
The Canonical Name record (CNAME) tells anyone visiting that subdomain to use the same DNS records as another domain / subdomain. If you are already using an A or AAAA record for a subdomain you wouldn’t use a CNAME. CNAME records must always point to another domain / subdomain, never directly to an IP address. For example, we can configure the CNAME of www.9tut.com points to 9tut.com so that they use the same IP address.
As opposed to forward DNS resolution (A and AAAA DNS records), the Reverse-lookup Pointer (PTR) record is used to look up domain names based on an IP address.
An SOA record or start of authority record specifies the DNS server providing authoritative information about an Internet domain, the email of the domain administrator, the domain serial number, and several timers relating
to refreshing the zone.
An example of a SOA record is shown below.
; name TTL class rr Nameserver email-address
mydomain.com. 14400 IN SOA ns.mynameserver.com. root.ns.mynameserver.com. (
2004123001 ; Serial number
86000 ; Refresh rate in seconds
7200 ; Update Retry in seconds
3600000 ; Expiry in seconds
600 ; minimum in seconds)
name - mydomain.com is the main name in this zone.
TTL - 14400 - TTL defines the duration in seconds that the record may be cached by client side programs. If it is set as 0, it indicates that the record should not be cached. The range is defined to be between 0 to 2147483647
(close to 68 years !) .
Class - IN - The class shows the type of record. IN equates to Internet. Other options are all historic. So as long as your DNS is on the Internet or Intranet, you must use IN.
Nameserver - ns.nameserver.com. - The nameserver is the server which holds the zone files. It can be either an
external server in which case, the entire domain name must be specified followed by a dot. In case it is defined
in this zone file, then it can be written as “ns” .
Email address - root.ns.nameserver.com. - This is the email of the domain name administrator. Now, this is really confusing, because people expect an @ to be in an email address. However in this case, email is sent to root@ns.nameserver.com, but written as root.ns.nameserver.com . And yes, remember to put the dot behind the
domain name.
Serial number - 2004123001 - This is a sort of a revision numbering system to show the changes made to the DNS Zone. This number has to increment , whenever any change is made to the Zone file. The standard convention is to use the date of update YYYYMMDDnn, where nn is a revision number in case more than one updates are done in a day. So if the first update done today would be 2005301200 and second update would be
2005301201.
Refresh - 86000 - This is time(in seconds) when the slave DNS server will refresh from the master. This value represents how often a secondary will poll the primary server to see if the serial number for the zone has increased (so it knows to request a new copy of the data for the zone). It can be written as “23h88M” indicating
23 hours and 88 minutes. If you have a regular Internet server, you can keep it between 6 to 24 hours.

Retry - 7200 - Now assume that a slave tried to contact the master server and failed to contact it because it was
down. The Retry value (time in seconds) will tell it when to get back. This value is not very important and can be a fraction of the refresh value.
Expiry - 3600000 - This is the time (in seconds) that a slave server will keep a cached zone file as valid, if it can’t contact the primary server. If this value were set to say 2 weeks (in seconds), what it means is that a slave
would still be able to give out domain information from its cached zone file for 2 weeks, without anyone knowing the difference. The recommended value is between 2 to 4 weeks.
Minimum - 600 - This is the default time(in seconds) that the slave servers should cache the Zone file. This is the most important time field in the SOA Record. If your DNS information keeps changing, keep it down to a day or less. Otherwise if your DNS record doesn’t change regularly, step it up between 1 to 5 days. The benefit of keeping this value high, is that your website speeds increase drastically as a result of reduced lookups. Caching servers around the globe would cache your records and this improves site performance.

Source : CCNA Training » Drag Drop Quiz 3.pdf | question 7 | page 8

Drag Drop Questions 3 | Question 08

Drag and drop the DHCP snooping terms from the left onto the descriptions on the right.

Éléments de réponse disponibles

1. snooping binding database
2. spurious DHCP server
3. DHCP server
4. untrusted
5. trusted

Réponses : associations du corrigé source

Élément / emplacement	Réponse associée
list of hosts on the network that are unknown to the administrative domain	snooping binding database
unknown DHCP server within an administrative domain	spurious DHCP server
network component that propagates IP addresses to hosts on the network	DHCP server
default state of all interfaces	untrusted
internal device under the control of the network administrator	trusted

Explication originale

+ list of hosts on the network that are unknown to the administrative domain: snooping binding database
+ unknown DHCP server within an administrative domain: spurious DHCP server
+ network component that propagates IP addresses to hosts on the network: DHCP server
+ default state of all interfaces: untrusted
+ internal device under the control of the network administrator: trusted

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